

CS 2614: Computer Organization

Lab 3

Spring 2025

Full adder

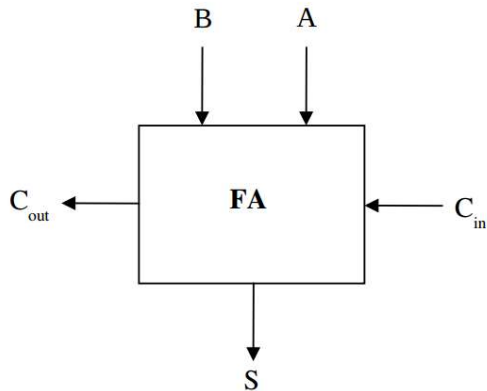
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Pre-Lab Exercise

1. (a) Derive the truth table for the full adder shown below. The truth table should have three inputs (A, B, and C_{in}) and two outputs (S and C_{out}).



Inputs			Outputs	
A	B	C_{in}	S	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

- (b) Use the K-Map approach to derive the simplified Boolean expressions for S and C_{out} .

K-map for S:

AB \ C_{in}	0	1
00	0	1
01	1	0
11	0	1
10	1	0

$$S = A \oplus B \oplus C_{in}$$

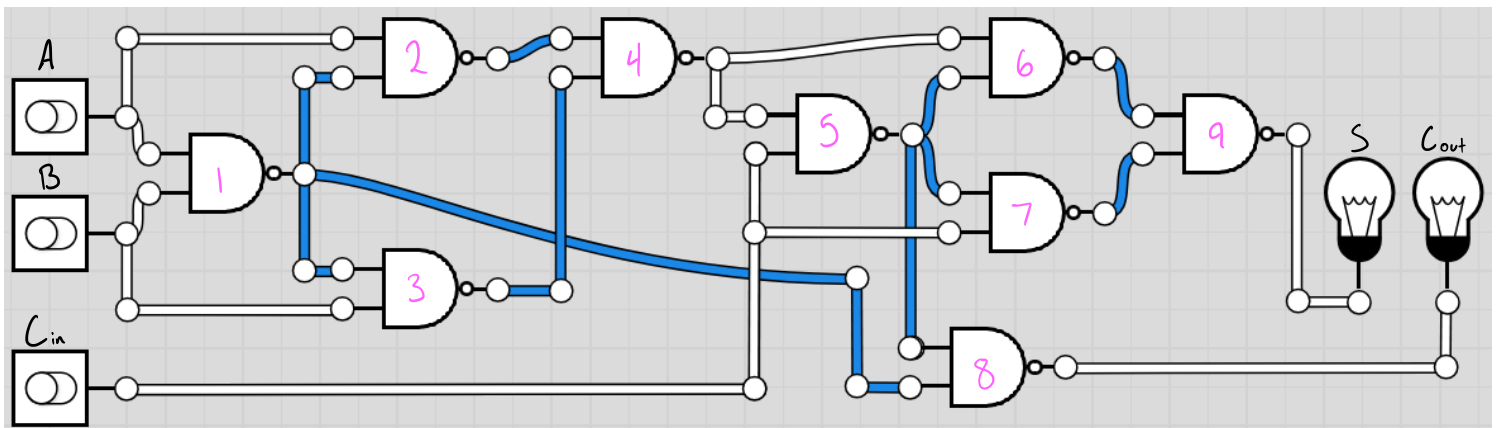
K-map for C_{out} :

AB \ C_{in}	0	1
00	0	0
01	0	1
11	1	1
10	0	1

$$C_{out} = AB + BC_{in} + AC_{in}$$

(c) Draw the logic diagram for the full adder circuit using NAND gates.

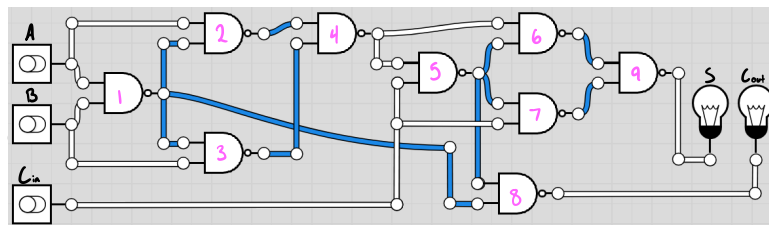
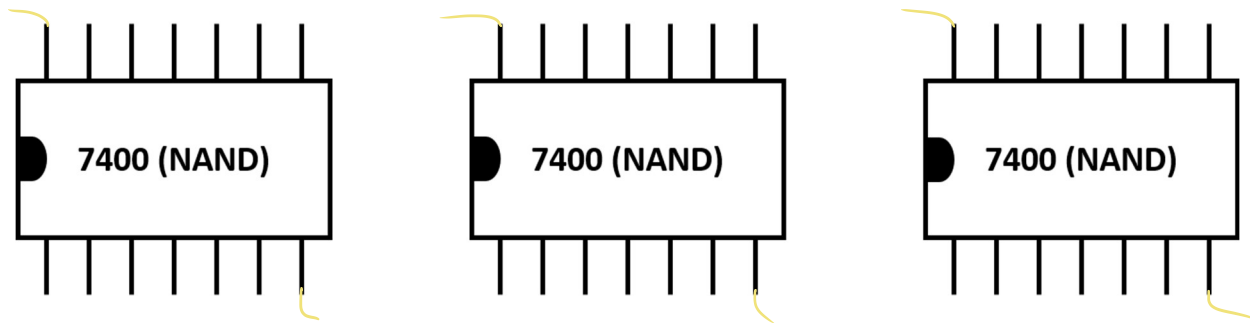
Inputs			Outputs	
A	B	C_{in}	S	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



Remember to upload this file on Canvas.

In-Lab Exercise

3. Draw the circuit from **Problem 1c** using the chips below (label each input/output).



4. Implement problem 3 on the breadboard